



Non-Markovian Cosmic-Ray Pitch-angle Transport from Mirror Interactions

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Abstract

Cosmic-ray pitch-angle transport in magnetohydrodynamic (MHD) turbulence is governed by the interplay between magnetic mirroring and gyroresonant scattering. We develop a guiding-center Langevin model with explicit mirror drift and gyroresonant diffusion to describe the pitch angle evolution. This model accurately captures our test-particle simulation results in three-dimensional MHD turbulence, driven both solenoidally and compressively. We find that magnetic mirroring can drive anomalous pitch-angle diffusion at large pitch angles (including 90°) with non-Markovian memory effects, which arise from the trapping of particles in magnetic wells. Gyroresonant scattering controls the escape rate from these wells. Across M_A , large-pitch-angle particles are jointly regulated by mirror trapping and gyroresonant escape, exhibiting a transition from anomalous to normal diffusive pitch-angle transport as scattering strengthens, whereas small-pitch-angle particles remain gyroresonance-dominated and diffusive throughout. The pitch-angle transport is found to be dominated by the compressible perturbations with marginal influence from Alfvén modes. In compressible turbulence with realistic damping accounted for, transit time damping treatment fully recovers mirror interactions.

Unified Astronomy Thesaurus concepts: [Cosmic ray astronomy \(324\)](#); [Cosmic rays \(329\)](#); [Magnetohydrodynamics \(1964\)](#); [Gamma-ray astronomy \(628\)](#)

1. Introduction

Cosmic-ray (CR) transport in magnetized turbulence underpins particle acceleration and nonthermal emission across astrophysical and space-plasma environments. In practice, CR propagation is often described as a diffusive process: it is assumed that pitch-angle scattering rapidly decorrelates particle directions, allowing transport to be modeled by a Markovian diffusion equation with well-defined coefficients (J. R. Jokipii 1966; R. Schlickeiser 2002). However, when scattering is intermittent or insufficient, particle trajectories retain memory of the underlying magnetic structure, rendering a diffusive description invalid (G. Zimbardo & S. Perri 2020; S. Maiti et al. 2022; J. Lübke et al. 2025; S. Zhao et al. 2025).

A key development beyond standard quasi-linear theory (QLT) was the recognition that near- 90° dynamics is strongly affected by compressive fluctuations and nonlinear mirror effects (G. M. Felice & R. M. Kulsrud 2001; H. Yan & A. Lazarian 2008). The δ -function resonance function of QLT yields vanishing pitch-angle scattering near 90° , and this limitation has long been understood as an artifact of the unperturbed guiding-center (GC) orbit assumption and the resulting idealized resonance function (J. R. Jokipii 1966; R. Kulsrud & W. P. Pearce 1969; G. Qin & A. Shalchi 2014). In particular, compressive fluctuations can induce bounce motion and transient trapping through conservation of the first adiabatic invariant $\mu_m = mv_\perp^2/(2B)$ or the resonant mirror interaction through the Landau resonance, the transit-time damping (TTD) process (C. J. Cesarsky & R. M. Kulsrud 1973; G. M. Felice & R. M. Kulsrud 2001; R. Schlickeiser 2002). The conservation of

magnetic moment naturally broadens the idealized QLT condition by a finite-width kernel, making TTD effective near $\mu \simeq 0$ and alleviating the 90° barrier (H. Yan & A. Lazarian 2008). Despite broad consensus on the importance of compressive structures, most existing approaches remain within a diffusive framework, in which mirror effects and compressions are typically incorporated as modified scattering rates within a Markovian transport picture. Recent studies have further renewed interest in the interplay between magnetic mirroring and pitch-angle scattering (A. R. Bell et al. 2025; V. López-Barquero & P. Desiati 2025; J. Lübke et al. 2025).

In this work, we demonstrate that mirror structures can drive a distinct regime, namely, mirror-induced anomalous diffusion that lies beyond both QLT and its nonlinear extensions. Rather than postulating a phenomenological memory kernel, we identify this behavior directly from first-principle test-particle dynamics in three-dimensional magnetohydrodynamic (MHD) turbulence. We introduce a GC Langevin model in which mirror interactions enter as a deterministic drift in μ , while gyroresonance is represented as a stochastic, Markovian diffusion term. This decomposition is intended to describe the regime in which mirror trapping and gyroresonance remain scale-separated. We show that this model accurately reproduces the test-particle results. At the ensemble level, the Langevin model corresponds to a focused Fokker–Planck equation with a spatially varying focusing term, extending constant-focusing treatments (J. A. Earl 1976; R. C. Tautz et al. 2014; J. Lasuik et al. 2017) to the multiscale, intermittent focusing, which is naturally produced by compressive turbulence. The combination of mirror effects and gyroresonant scattering has two linked consequences that regulate the global transport. First, it enables particles to traverse $\mu \simeq 0$ (90° barrier) via a deterministic drift. Second, because this drift within a local well is reversible and weakly accumulative, it



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can also confine particles in intermittent mirror wells, yielding heavy-tailed trapping times and thus non-Markovian, anomalous pitch-angle evolution. In this framework, particle transport is controlled by the competition between mirror trapping and gyroresonant decorrelation. By directly linking anomalous transport to a concrete microscopic mechanism, we aim to identify the conditions (e.g., Alfvén Mach number M_A) under which diffusive descriptions are acceptable and explicit non-Markovian effects must be considered. It offers microscopic insights into CR transport across scales, from early-time propagation after escaping CR accelerators (J. A. Earl 1974; A. Y. Prosekin et al. 2015; S. Recchia et al. 2021) to the formation of spatially extended nonthermal emission around accelerators (A. Abeysekara et al. 2017; R.-Y. Liu et al. 2019; F. Aharonian et al. 2021; Lhaaso Collaboration 2024).

2. Method

2.1. MHD Turbulence

We generate 3D isothermal ideal-MHD turbulence using the ATHENA++ code within a triply periodic box of size L at a resolution of $N^3 = 1024^3$ (J. M. Stone et al. 2020). Energy is injected at the scale $k_{\text{inj}} = 2$. Turbulence is continuously driven by an Ornstein–Uhlenbeck process $\mathbf{F}^{\text{turb}}$, where the solenoidal/compressive mixture is controlled via the projection operator $P_{ij}^\zeta = \zeta P_{ij}^\perp + (1 - \zeta) P_{ij}^\parallel$ (K. D. Makwana & H. Yan 2020; P. Kempfski et al. 2025). We set $\zeta = 0.1$ for compressive driving data cubes, and $\zeta = 1$ for solenoidal driving data cubes. We adopt a plasma beta of $\beta = 0.5$. The Alfvén Mach number M_A is varied by adjusting the energy injection rate. The corresponding sonic Mach number is given by $M_s = M_A \sqrt{2/(\hat{\gamma}\beta)} = 2M_A$ for isothermal plasma ($\hat{\gamma} = 1$). In each run, data are collected only after the turbulence reaches a statistically steady state.

2.2. Test-particle Integration

We trace CR trajectories by integrating the motion of relativistic test particles in the 3D MHD turbulence using the Boris push method with periodic boundaries (W. H. Press et al. 2007). Since the speed of a relativistic particle greatly exceeds the Alfvén speed, the magnetic field is treated as stationary, and the induced electric field is neglected (A. Beresnyak et al. 2011; S. Xu & H. Yan 2013; R. Cohet & A. Marcowith 2016; S. Maiti et al. 2022). The particle motion follows the Lorentz force equation. The gyrofrequency and Larmor radius are $\Omega_0 = qB_0/(\gamma mc)$ and $r_L = \gamma mvc/(qB_0)$, respectively. q is the particle charge. m is the rest mass. c is the speed of light. v is the particle speed. γ is the Lorentz factor. B_0 is the mean magnetic field. We set $r_L = 5 l_{\text{grid}}$ in default, where l_{grid} is an MHD-normalized unit of the grid spacing. This choice places the corresponding resonant wavenumber at $k = N/5/(2\pi) \approx 32$, which is well within the inertial range of the turbulence (Appendix A Figure 5).

2.3. Guiding-center Langevin Model

To disentangle mirror effects from gyroresonant scattering, we construct a GC Langevin model that describes the evolution of the pitch-angle cosine μ as a drift–diffusion

process (M. Zhang 1999), i.e.,

$$d\mu = -\frac{1 - \mu^2}{2B} v \nabla_s B dt + \frac{\partial D_{\mu\mu}^G}{\partial \mu} dt + \sqrt{2D_{\mu\mu}^G(\mu)} \eta_t, \quad (1)$$

where $\nabla_s = \hat{\mathbf{b}} \cdot \nabla$ (with $\hat{\mathbf{b}} = \mathbf{B}/B$) denotes the gradient along the local magnetic field, and η_t is a Markovian and Gaussian white noise satisfying $\langle \eta(t_1) \eta^*(t_2) \rangle = \delta(t_1 - t_2) \langle \eta(t) \eta^*(t) \rangle$, and $\langle \eta(t) \rangle = 0$, $\langle \eta(t) \eta^*(t) \rangle = dt$ (X. Zhang et al. 2020; J. Lübke et al. 2025). The first term represents the deterministic mirror (focusing) drift in the adiabatic regime, where the magnetic-field variation scale satisfies $\ell \gtrsim r_g$. Micromirror structures with $\ell \ll r_g$, such as those produced by kinetic instabilities in high- β plasma and possibly relevant in intracluster medium-like environments (e.g., P. Reichherzer et al. 2025), are not included in either our simulations or our theoretical analysis. The last two terms together represent stochastic gyroresonant scattering, characterized by the diffusion coefficient $D_{\mu\mu}^G$, which is computed following Equation (5) of H. Yan & A. Lazarian (2008) by integrating the turbulence spectrum over nonlinearly broadened cyclotron-resonant scales,

$$D_{\mu\mu}^G = \frac{\Omega_0^2(1 - \mu^2)}{B_0^2} \int d^3k \sum_{n \geq 1} R_n(k_\parallel v_\parallel - \omega \pm n\Omega_0) \times \left[I^A(\mathbf{k}) \frac{n^2 J_n^2(W)}{W^2} + \frac{k_\parallel^2}{k^2} J_n'^2(W) I^M(\mathbf{k}) \right], \quad (2)$$

where $v_\parallel$ ($v_\perp$) is the particle speed parallel (perpendicular) to B_0 , $k_\parallel$ ($k_\perp$) is the parallel (perpendicular) wavevector, J_n (J_n') is the n th-order Bessel function (and its derivative), R_n is the n th-order resonant function, $W = k_\perp v_\perp / \Omega_0$, and I^A and I^M are the scalar energy spectra for Alfvénic and magnetosonic modes, respectively.

The GC position $\mathbf{R}$ evolves as (G. G. Howes et al. 2006)

$$d\mathbf{R} = v_\parallel \frac{\mathbf{B}}{B} dt - \frac{c \mu_m}{q} \frac{\nabla B \times \mathbf{B}}{B^2} dt, \quad (3)$$

where $\mu_m = mv_\perp^2/(2B)$ is the magnetic moment. $v_\parallel$ ($v_\perp$) is the particle speed parallel (perpendicular) to $\mathbf{B}$. By integrating the coupled Langevin Equations (1) and (3), we advance the GC trajectories in the 3D MHD turbulence fields.

3. Results

3.1. Separating Mirror Interactions from Gyroresonance

We characterize pitch-angle transport using the ensemble spread of the pitch-angle cosine, $\langle |\Delta\mu(t)| \rangle = \langle |\mu(t + t_0) - \mu(t_0)| \rangle$, where t_0 is the reference start time (by default $t_0 = 0$; $\mu_0 \equiv \mu(0)$). We quantify its scaling $\langle \Delta\mu \rangle \propto t^\alpha$. Here, $\alpha = 0.5$ corresponds to normal diffusion, $\alpha < 0.5$ to subdiffusion, and $\alpha > 0.5$ to superdiffusion (with $\alpha = 1$ indicating ballistic growth). Time is normalized to the gyroperiod t_g . To isolate the physical channels that drive μ evolution, we exploit the orthogonality of Alfvénic and compressible perturbations in $\mathbf{k}$ -space and decompose the turbulence into Alfvénic (A) and compressible (C) components (J. Cho & A. Lazarian 2002, 2003).

In the C component, mirror interactions and gyroresonance coexist. Mirror is controlled by $\nabla|B|$ and is inherently broadband across the inertial range, whereas gyroresonance is concentrated near cyclotron-resonant scales. Notably, mirror

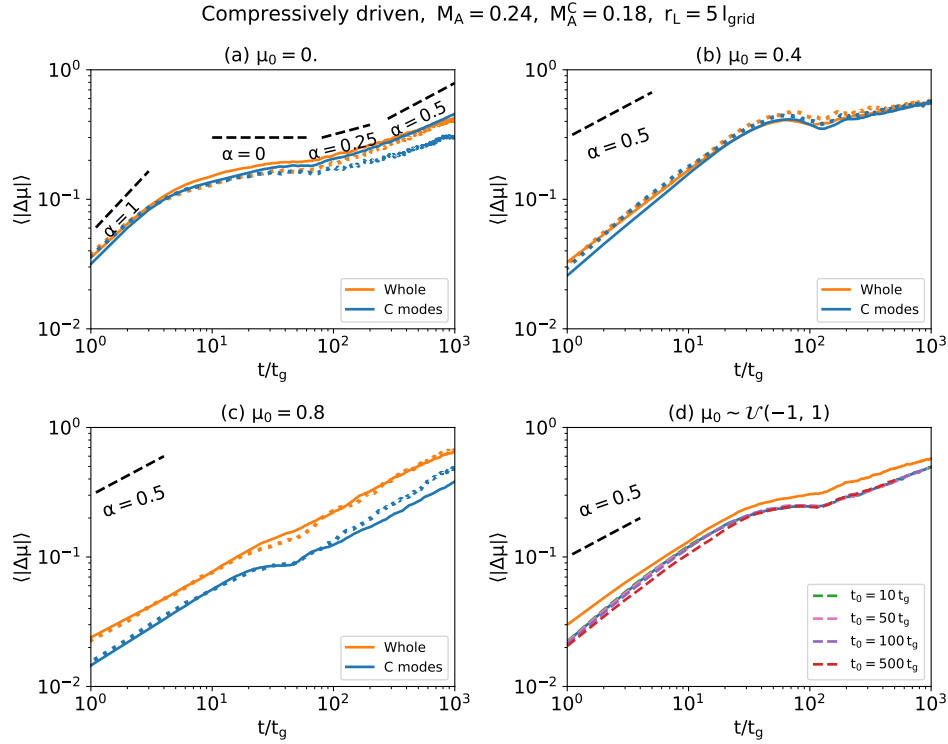


Figure 1. Time evolution of the mean pitch-angle increment $\langle |\Delta\mu(t)| \rangle$ for particles with different initial μ_0 in compressively driven turbulence of $M_A^C = 0.18$, $M_A = 0.24$, with $r_L = 5 l_{\text{grid}}$. Solid lines show the test-particle results, while dotted lines show the GC Langevin results. (a) Initial pitch angle $\mu_0 = 0$, (b) $\mu_0 = 0.4$, (c) $\mu_0 = 0.8$, (d) μ_0 is sampled from a uniform distribution $\mathcal{U}(-1, 1)$. Colored dashed lines showcase different reference start time t_0 .

interactions do not rely on a specific resonant scale and exhibit no intrinsic rigidity dependence (Equation (1)), so varying the Larmor radius primarily affects gyroresonant scattering. To construct mirror-dominated reference cases, we suppress gyroresonance by artificially reducing the Larmor radius to a sufficiently small value (e.g., $r_L = 1 l_{\text{grid}}$), thereby shifting the resonant scale into the dissipation range. Comparing results from the full turbulence, the mode-decomposed fields, and these gyroresonance-suppressed runs allows us to disentangle deterministic mirror confinement from stochastic resonant decorrelation.

In the A component, the perturbations are predominantly transverse to the mean field $\mathbf{B}_0$. As a result, the mirror-relevant gradients $\nabla|B|$ enter only at higher order. Mirror/TTD effects therefore vanish at leading order, and the A-modes contribution to pitch-angle scattering is dominated by gyroresonance. However, gyroresonant scattering by Alfvénic turbulence is inefficient near $\mu \simeq 0$. Indeed, we see that in large pitch angle cases (small μ_0) the μ evolution resulting from the C component is nearly identical to that from the full turbulence (Figures 1(a) and (b)).⁵ This implies that the A component contributes negligibly in this regime. This behavior is further strengthened by the scale-dependent anisotropy of Alfvénic turbulence, which suppresses power at the parallel wavenumbers $k_{\parallel}$ required for the resonance with particles in a realistic astrophysical environment (P. Goldreich & S. Sridhar 1995; H. Yan & A. Lazarian 2002, 2004).

Another question is whether field-line curvature or Richardson superdiffusion of A modes could provide an additional contribution (A. Lazarian & H. Yan 2014; S. Maiti et al. 2022). To assess the combined effect of field-line curvature and mirror interactions, we performed a simulation using the solenoidally driven datacube with $r_L = 1 l_{\text{grid}}$. We see the C modes and whole datacube result in nearly identical $\langle |\Delta\mu| \rangle$ behavior (Figures 2(a) and (b)), indicating that A modes field line curvatures are also negligible.

3.2. Low- M_A : Compound (Multistage) Pitch-angle Transport

Figure 1 shows $\langle |\Delta\mu| \rangle$ for a sub-Alfvénic compressible run ($M_A^C = 0.18$ for the C components, $M_A = 0.24$ for the whole datacube) with different initial μ_0 . For each μ_0 , we inject an ensemble of 5000 particles randomly throughout the turbulence volume. This setup is not intended to model a specific injection scenario. Instead, it serves as a diagnostic of the μ -dependent pitch-angle evolution, noting that particles span a broad range of μ at any time. A clear regime dependence emerges. Particles injected near $\mu_0 = 0$ (Figure 1(a)) do not enter normal diffusion immediately, but exhibit a four-stage evolution with successive scaling exponents $\alpha \simeq 1, 0, 0.25$ and 0.5 . This sequence reflects the competition between conservative mirror confinement and weak gyroresonant decorrelation. At early times, coherent mirror forcing produces an apparent ballistic growth ($\alpha \simeq 1$), followed by a plateau ($\alpha \simeq 0$) associated with repeated reflections and confinement within a narrow μ band. At intermediate times, rare resonant kicks gradually accumulate and erode phase coherence of trapped orbits, leading to subdiffusion ($\alpha < 0.5$). Only at a later time, after particles escape local mirror wells and

⁵ We do not present the scattering results for pure Alfvénic modes, as the lack of mirror interaction and the Alfvénic spectral anisotropy leads to inefficient pitch-angle scattering, rendering the results sensitive to numerical integration accuracy.

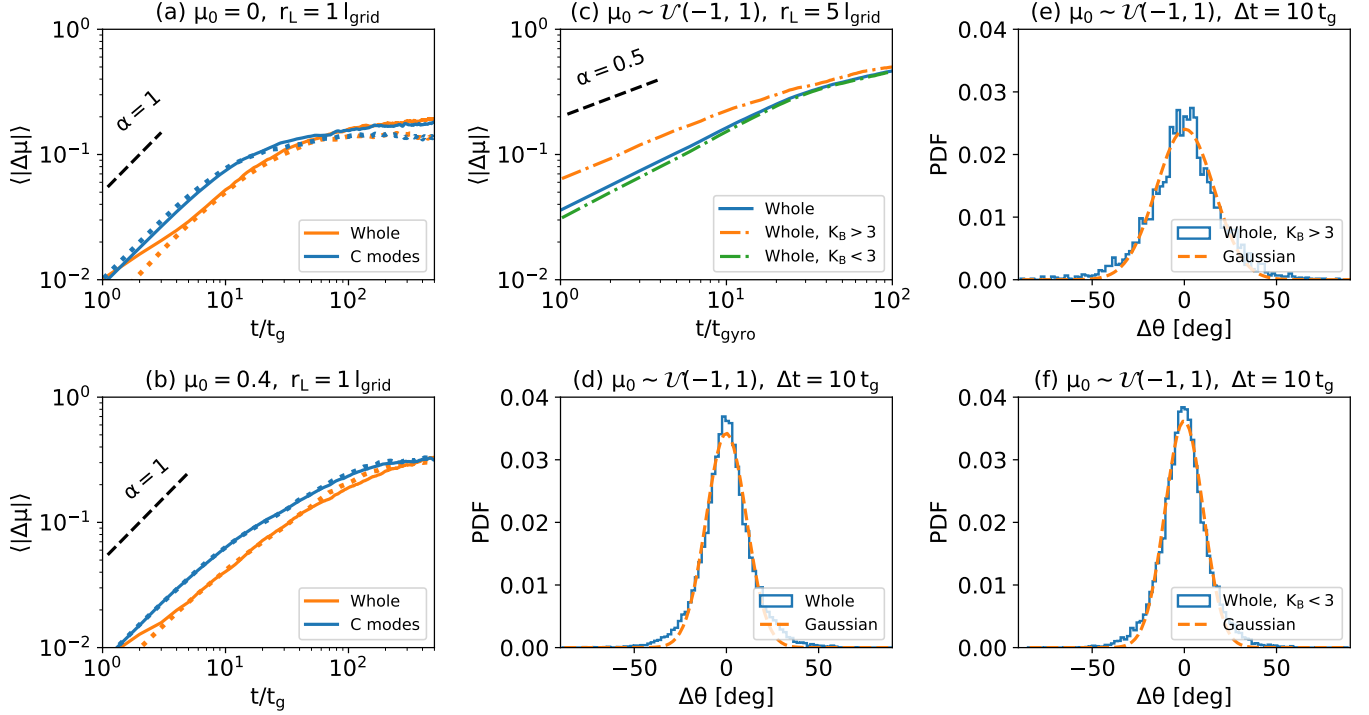
Solenoidally driven, $M_A = 0.33$, $M_A^C = 0.16$ 

Figure 2. Statistics of particles in solenoidally driven turbulence of $M_A = 0.33$, $M_A^C = 0.16$. (a) Time evolution of the mean pitch-angle increment $\langle |\Delta\mu(t)| \rangle$ of particles with $\mu_0 = 0$ and $r_L = 1 l_{\text{grid}}$, so as to suppress gyroresonance and isolate the effects of magnetic mirroring and field-line curvature. Solid lines show the test-particle results, while dotted lines show the GC Langevin results. $\langle \Delta\mu \rangle$ rises faster initially in the C modes than in the whole datacube, because A modes increase $|B|$ without contributing to $\nabla_\perp B$, thus slightly reducing mirroring $\propto \nabla_\perp B / |B|$. (b) Same as panel (a), but for $\mu_0 = 0.4$. (c) $\langle |\Delta\mu(t)| \rangle$ of particles with μ_0 isotropically injected from a uniform distribution $\mathcal{U}(-1, 1)$ and $r_L = 5 l_{\text{grid}}$. Dashed-dotted lines show results of particles injected at high-kurtosis regions and low-kurtosis regions separately. (d) Probability density functions (pdfs) of pitch angle increment $\Delta\theta$ of test-particle simulations for $\mu_0 \sim \mathcal{U}(-1, 1)$, $\Delta t = 10 t_g$. The dashed-dotted curve shows the best-fit Gaussian distribution. (e) Same as panel (d), but for particles injected at high-kurtosis regions. (f) Same as panel (d), but for particles injected at low-kurtosis regions.

substantially deviate from their initial μ_0 , $\langle \Delta\mu \rangle$ approaches two normal diffusion ($\alpha \simeq 0.5$). Mirror interactions therefore play a dual role. They enable particles to cross the 90° barrier via drift, moving them out of the 90° inefficient scattering regime and thereby enhancing gyroresonant scattering and escape, while also producing intermittent trapping that delays the onset of the diffusive regime.

This interpretation is supported by the gyroresonance-suppressed runs: the early ballistic and plateau phases closely track the gyroresonance-suppressed case (Appendix A Figure 6), indicating that they are controlled primarily by mirror dynamics, while the subsequent transition from subdiffusion to diffusion requires the cumulative effect of gyroresonant scattering. Thus, in the low- M_A regime, escape from mirror potentials naturally produces anomalous scaling over extended times before the diffusive limit is reached.

In contrast, particles injected at small pitch angle, $\mu_0 = 0.8$ (Figure 1(c)), exhibit normal diffusion throughout the entire evolution. Nevertheless, non-Markovianity at large pitch angles can regulate the global transport, because particle pitch angles evolve dynamically and undergo repeated trapping and escape episodes. At $\mu_0 = 0.4$, the mirror-induced anomalous transport already becomes pronounced, as shown in Figure 1(b). Therefore, an anomalous diffusion regime can also be seen in the evolution of $\langle |\Delta\mu(t)| \rangle$ for an isotropic pitch-angle distribution (Figure 1(d)).

To quantify the non-Markovian nature of pitch-angle transport, we further analyze the statistical distribution of the

pitch-angle increments $\Delta\mu = \mu(\Delta t) - \mu_0$ with two representative cases, $\mu_0 = 0$ and $\mu_0 = 0.8$ (Figure 3). For $\mu_0 = 0$, the increment probability density functions (pdfs) at $\Delta t = 10 t_g$ exhibits leptokurtic and are poorly described by a single Gaussian. It is well-fit by a stretched Gaussian (R. Kupferman 2004),

$$P(\Delta\mu) = \frac{\kappa}{2\sigma \Gamma(1/\kappa)} \exp \left[- \left(\frac{|\Delta\mu|}{\sigma} \right)^\kappa \right], \quad (4)$$

where σ sets the characteristic width and κ controls the tail weight ($\kappa = 2$ recovers a Gaussian; $\kappa < 2$ yields leptokurtic). For the $\mu_0 = 0$ case, we obtain the best-fit shape parameter $\kappa \simeq 0.72$ (Figure 3(a)), indicating non-Gaussian increments stemming from dynamics associated with mirror trapping and rare escape events. In contrast, for $\mu_0 = 0.8$ the increment pdf is close to Gaussian (Figure 3(b)), indicating rapid decorrelation and approximately Markovian diffusion.

To further quantify the temporal statistics, we measure the first-passage time Δt required for $|\Delta\mu|$ to reach a threshold $|\Delta\mu| = 0.05$, which serves as a representative increment rather than a unique choice. The shape of the tail encodes the underlying transport regime. For $\mu_0 = 0$, the waiting-time distribution exhibits a power-law tail (Figure 3(c)), implying a broad distribution of residence times and a non-Markovian process driven by localized mirror trapping with sporadic gyroresonant escape. For $\mu_0 = 0.8$, the distribution shows a

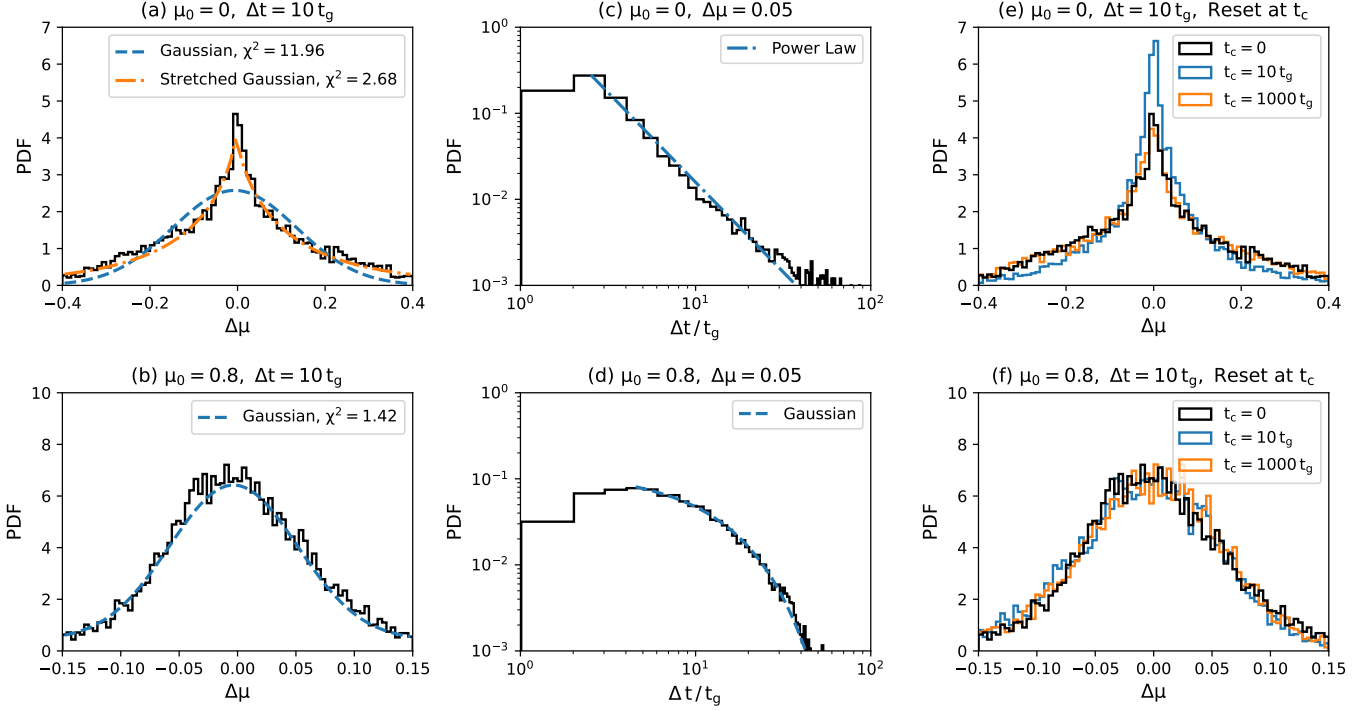
Compressively driven, $M_A^C = 0.18$, $r_L = 5 l_{\text{grid}}$ 

Figure 3. Statistical properties of $\Delta\mu$ measured from test-particle simulations in a sub-Alfvénic compressive turbulence with $M_A^C = 0.18$. (a) Probability density functions (pdfs) of pitch angle cosine increment $\Delta\mu$ of test-particle simulations for $\mu_0 = 0$, $\Delta t = 10 t_g$. The dashed and dashed-dotted curves show fits with a stretched Gaussian and a Gaussian distribution, respectively. The legend lists the reduced chi-square χ^2 for each fit. (b) Same as (a), but for $\mu_0 = 0.8$. (c) Waiting-time distribution for pitch angle cosine increment reaches a threshold $|\Delta\mu| = 0.05$ for $\mu_0 = 0$. (d) Same as (b), but for $\mu_0 = 0.8$. (e) The pdf of $\Delta\mu$ for $\mu_0 = 0$ and $\Delta t = 10 t_g$, computed after resetting $\mu(t_c) = \mu_0$ at different reference times t_c , i.e., $\Delta\mu = \mu(t_c + \Delta t) - \mu(t_c)$. (f) Same as (e), but for $\mu_0 = 0.8$.

rapid cutoff (Figure 3(d)), consistent with the Markovian process.

Memory effects are probed directly by “reset” experiments: at a reference time t_c , we reset $\mu(t)$ to the injected value μ_0 while keeping the spatial position unchanged, and then remeasure the increment statistics $\Delta\mu = \mu(\Delta t + t_c) - \mu_0$. For $\mu_0 = 0$, the pdf depends strongly on t_c at early times (Figure 3(e)) and relaxes only after long evolution ($t_c \sim 10^3 t_g$). This behavior reflects mirror-induced spatial focusing: particles initially distributed randomly are rapidly trapped in localized magnetic wells, and only escape after accumulating sufficient gyroresonant scattering to restore spatial randomness. In contrast, for $\mu_0 = 0.8$, the pdfs collapse rapidly for different t_c (Figure 3(f)), demonstrating fast decorrelation. Together, these diagnostics identify a non-Markovian, anomalous diffusion regime at low M_A .

3.3. High- M_A : Rapid Decorrelation and Diffusive Transport

At high- M_A , the pitch-angle dynamics remain dominated by the competition between intermittent mirror trapping and gyroresonant escape. Yet gyroresonant scattering becomes sufficiently efficient that most particles rapidly escape local magnetic potential wells. Within a few gyroperiods, the pitch-angle distribution becomes randomized, and the memory of the injected μ_0 is quickly lost. This holds even for $\mu_0 = 0$, due to broadened/nonlinear gyroresonance effects (H. Yan & A. Lazarian 2008) and to the stronger mirror-induced pitch-angle drift, which can sweep particles into regimes where gyroresonant scattering is more effective. While non-Markovianity can still occur for a subpopulation of particles at large

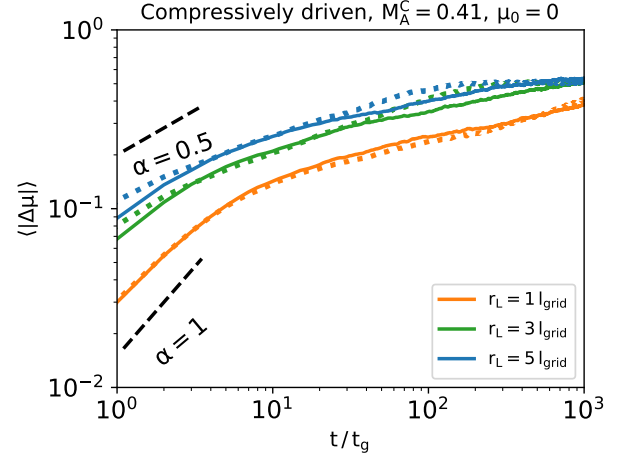


Figure 4. Time evolution of the mean pitch-angle increment $\langle |\Delta\mu(t)| \rangle$ for particles with initial $\mu_0 = 0$ in compressively driven turbulence of $M_A^C = 0.41$. Solid lines show the test-particle results, while dotted lines show the GC Langevin results. The gyroperiod t_g depends on Larmor radii R_L , and thus varies for different particle populations here.

pitch angles, rapid escape and efficient decorrelation result in the ensemble-averaged $\langle \Delta\mu \rangle$ dynamics transitioning to a diffusive scaling $\langle \Delta\mu \rangle \propto t^\alpha$ with $\alpha \simeq 0.5$ (red curves in Figure 4).

To further disentangle gyroresonance from mirror effects, we vary the strength of gyroresonant scattering by changing the particle Larmor radius. Our benchmark choice is $r_L = 5 l_{\text{grid}}$, for which the resonant wavenumbers lie in the inertial range and gyroresonance is well resolved. We then

progressively reduce the Larmor radius to $r_L = 3$ and $1 l_{\text{grid}}$, shifting the resonance toward smaller scales and effectively suppressing gyroresonant interactions. As shown in Figure 1(f), when r_L is sufficiently small, $\langle \Delta \mu \rangle$ departs from the diffusive law and exhibits anomalous scaling. This controlled simulation demonstrates that the anomalous regime is rooted in prolonged mirror trapping with inefficient resonant escape. When particles escape from local mirror wells quickly, they traverse multiple stochastic mirror structures, and the cumulative effect of these encounters enhances scattering and accelerates decorrelation. In this regime, mirror interactions mainly act as an additional broadband scattering enhancement, and the transport becomes effectively diffusive.

In the full diffusive limit, we can start from the mirror force update in Equation (1) and evaluate $D_{\mu\mu}$ via the Taylor–Green–Kubo (TGK) formula (R. Kubo 1957; A. Shalchi et al. 2004). If we restrict to GC trajectories and overlook the fast gyromotion, then the orbit phase factor reduces to $\exp(ik_{\parallel} v \mu t)$. In this case, there is no dependence on the gyro-orbit. The mirror force only couples to gradients of $B_{\parallel}$ along the field line, and thus does not require information about the gyrophase. Accordingly, only the $n = 0$ resonance survives, and no Bessel functions appear. The $D_{\mu\mu}^{\text{Mirror}}$ is then calculated as

$$D_{\mu\mu}^{\text{Mirror}} = \frac{v^2(1 - \mu^2)^2}{4B_0^2} \int d^3k k_{\parallel}^2 P_{\parallel}^{(M)}(\mathbf{k}) R_0, \quad (5)$$

as detailed in Appendix C. While the standard TTD contribution to pitch-angle diffusion is written as (see H. J. Voelk 1975; H. Yan & A. Lazarian 2008)

$$D_{\mu\mu}^{\text{TTD}} = \frac{\Omega^2(1 - \mu^2)}{B_0^2} \int d^3k R_0 \frac{k_{\parallel}^2}{k^2} [J_0'(W)]^2 I^{(M)}(\mathbf{k}). \quad (6)$$

In the regime of $W \ll 1$, the Bessel functions can be expanded as $|J_0'(W)| \simeq \frac{W}{2}$ so that the TTD form reduces exactly to the GC mirror result

$$\begin{aligned} D_{\mu\mu}^{\text{TTD}} &\xrightarrow{W \ll 1} \frac{\Omega_0^2(1 - \mu^2)}{B_0^2} \int d^3k R_0 \frac{k_{\parallel}^2}{k^2} \left(\frac{W}{2}\right)^2 I^{(M)}(\mathbf{k}) \\ &= \frac{v_{\perp}^2(1 - \mu^2)}{4B_0^2} \int d^3k k_{\parallel}^2 R_0 \frac{k_{\perp}^2}{k^2} I^{(M)}(\mathbf{k}) \\ &= \frac{v^2(1 - \mu^2)^2}{4B_0^2} \int d^3k k_{\parallel}^2 R_0 P_{\parallel}^{(M)}(\mathbf{k}) = D_{\mu\mu}^{\text{Mirror}}, \end{aligned} \quad (7)$$

where $P_{\parallel}^M$ is the parallel magnetic-power spectrum, and I^M denotes the scalar magnetic-power spectrum. Thus, the GC mirror interpretation is independent of the detailed gyro-orbit, sensitive to all compressive $\delta B_{\parallel}$ fluctuations, and equivalent to TTD in the small $k_{\perp}$ limit, which corresponds to either long wavelength or quasi-slab modes. The latter is expected for fast modes in low- β plasmas due to collisionless damping (H. Yan & A. Lazarian 2004, 2008; V. Petrosian et al. 2006; T. K. Suzuki et al. 2006; S. Zhao et al. 2024; C. Hou et al. 2025; S. Zhao et al. 2026).

3.4. Mean Free Path from $D_{\mu\mu}$ and $D_{\parallel}$

The parallel mean free path $\lambda_{\parallel}$ can be estimated from pitch-angle diffusion under the assumption of an isotropic

Table 1 Mean Free Path $\lambda_{\parallel}$ Inferred from Two Diagnostics of $D_{\mu\mu}$ and $D_{\parallel}$			
	$M_A = 0.51$	$M_A = 0.58$	$M_A = 0.66$
$\lambda_{\parallel} [L_{\text{box}}]$ from $D_{\parallel}$	1.09	0.69	0.50
$\lambda_{\parallel} [L_{\text{box}}]$ from $D_{\mu\mu}$	0.96	0.80	0.55

pitch-angle distribution, using the following expression (J. A. Earl 1974; S. Maiti et al. 2022):

$$\lambda_{\parallel} = \frac{3v}{4} \int_0^1 \frac{(1 - \mu^2)^2}{D_{\mu\mu}} d\mu, \quad D_{\mu\mu} = \frac{\langle (\mu - \mu_0)^2 \rangle}{2t}. \quad (8)$$

Note that $D_{\mu\mu}$ here denotes an effective pitch-angle diffusion coefficient that incorporates the combined effects of both gyroresonance and mirror interactions.

Alternatively, the mean free path can also be estimated from the spatial diffusion coefficient (J. A. Earl 1974; S. Maiti et al. 2022):

$$\lambda_{\parallel} = \frac{3D_{\parallel}}{c}, \quad D_{\parallel} = \frac{\langle (z - z_0)^2 \rangle}{2t}. \quad (9)$$

These two methods are expected to agree when pitch-angle scattering is efficient over the entire μ domain, so that the distribution of the pitch angle $f(\mu)$ relaxes to isotropic.

In our simulations, this equivalence breaks down whenever scattering is incomplete, such as in the pure Alfvénic modes or in the low- M_A datacube cases. In these regimes, particles with large pitch angles cannot be sufficiently scattered and do not reach the normal diffusive regime. In this case, a stationary $D_{\mu\mu}$ and $\lambda_{\parallel}$ cannot be obtained reliably.

When scattering is sufficient, most notably in the high- M_A data cubes, the two estimates of mean free path converge. In this regime, particles quickly drift and escape individual mirror potential wells and decorrelate from the initial state. Pitch-angle scattering behavior transitions to a normal diffusion regime within a short time. Mirror/TTD interactions then contribute efficiently to a well-defined $D_{\mu\mu}$, and Equations (8) and (9) yield consistent values of the parallel mean free path (Table 1).

3.5. Large-angle Scattering from Intermittency

Some studies have suggested that scattering may be governed by high-order intermittency, which produces rare but large-angle scattering events (e.g., I. S. Butsky et al. 2024). To assess the role of higher-order intermittency, we disentangle contributions from coherent structures and the background cascade using second- and fourth-order structure functions of magnetic increments. Following the procedure in C. Hou et al. (2025), we quantify magnetic intermittency through the kurtosis

$$K_B(\Delta \mathbf{r}) \equiv \frac{\langle [\Delta B]^4 \rangle}{\langle [\Delta B]^2 \rangle^2}, \quad (10)$$

where $\Delta B \equiv B(\mathbf{r} + \Delta \mathbf{r}) - B(\mathbf{r})$ and $\Delta \mathbf{r}$ is the spatial separation vector. Values $K_B > 3$ indicate non-Gaussian, heavy-tailed statistics typical of intermittent structures. We inject particle ensembles into high- K_B and low- K_B regions separately with isotropically distributed pitch angles and measure the $\langle |\Delta \mu| \rangle$ behavior. As shown in Figures 2(c)–(f), particles initialized in high- K_B regions exhibit larger $\langle |\Delta \mu| \rangle$, but these regions occupy a small filling factor (regions with $K_B > 3$ occupy about 12.93% of the total simulation volume),

and the overall $\langle |\Delta\mu| \rangle$ is not dominated by intermittent hotspots, indicating that scattering is governed by broadly distributed, lower-order fluctuations rather than rare intermittency.

4. Discussion

4.1. Negligence of Damping Effects

In MHD turbulence generated by Athena++, artificial isotropic viscosity is added at small-scale dissipation ranges to maintain numerical stability and to prevent energy pile-up at large k (J. M. Stone et al. 2020). In a real plasma environment, multiple physical damping channels operate, and their relative importance depends on both wavelength scale and plasma parameters. Because interstellar phases differ in density, ionization fraction, and plasma β , they have different dominant damping mechanisms (H. Yan & A. Lazarian 2004; H. Yan et al. 2004). Low-density rarefied media such as the solar wind and Galactic halo tend to be governed by collisionless damping. In warm ionized or neutral media, collisional viscosity and thermal conduction gain importance. In partially ionized phases, ion–neutral friction preferentially damps currents transverse to $\mathbf{B}$. In all cases, damping preferentially suppresses large- k power. Moreover, for compressive (mostly fast-mode) fluctuations in low- β plasmas, both theory and recent simulations and observations show stronger damping at oblique/quasi-perpendicular angles than in quasi-slab geometries (V. Petrosian et al. 2006; H. Yan & A. Lazarian 2008; S. Zhao et al. 2024; C. Hou et al. 2025; H. Yan et al. 2026; S. Zhao et al. 2026). Including physical damping therefore further suppresses the high- k portion of the spectrum and quasi-perpendicular components, and gradually restores the equivalence between TTD and mirror interactions in the diffusive limit.

4.2. Anisotropic Particle Distribution and Gyroresonance Instability

In this paper, the possible kinetic feedback is not included. In the case of low- M_A turbulence, for instance, the insufficient scattering is bound to drive anisotropy in the particle distribution. The pressure anisotropy can excite gyroresonance instability, which can increase the scattering rate, in turn (H. Yan & A. Lazarian 2011; O. Lebiga et al. 2018). As a result, particle transport can restore normal diffusion through the self-excited Alfvén waves (see R. M. Kulsrud 2005; H. Yan & A. Lazarian 2012; A. M. Bykov et al. 2013). The interplay between mirror interaction and the gyroresonance counting for the feedback will be studied elsewhere.

5. Summary

In this work, we showed that mirror structures in turbulent magnetic fields drive non-Markovian pitch-angle diffusion at large pitch angles, which influences the transport picture of CRs. By combining full-orbit test-particle simulations in 3D MHD turbulence with a GC Langevin model, we disentangle the roles of gyroresonant scattering and mirror interactions. We find that mirror interactions act multiscale: they are local in real space, generate non-Markovian memory through trapping in mirror potential wells, and statistically produce anomalous pitch-angle transport. At large Alfvénic Mach number M_A , gyroresonant kicks become strong enough to let particles hop across many localized mirror structures, causing

decoherence and restoring to normal diffusion. The coupling between these two processes governs the transport of particles. Our main results can be summarized as follows:

1. Mirror structures, generated primarily by compressive modes (and by pseudo-Alfvén fluctuations in the incompressible limit), introduce a deterministic drift in μ which naturally breaks the QLT prediction of pitch-angle scattering and brings the particles out of the 90° barrier.
2. Gyroresonant scattering modulates the escape from mirror wells and is essential for establishing normal diffusion. The higher the Alfvén Mach number M_A is, the faster normal diffusion restores.
3. The mirror effect leads to non-Markovian, anomalous pitch-angle diffusion with memory, characterized by platykurtic distributions of pitch-angle increment and a heavy-tailed pdf of the waiting time.
4. The pitch-angle transport is dominated by the interaction with compressible fluctuations. The mean free path calculated with $D_{\mu\mu}$ contributed from the compressible modes agrees well with the measurement from spatial displacement. The contributions from Alfvén modes and large-angle scattering from the intermittent structures are negligible.
5. In the diffusive limit, TTD treatment is adequate for capturing the magnetic mirror interaction in the case of k , or $k_\perp \ll 1/r_L$.

Our results suggest that CR propagation experiences intermittent spatial trapping of particles in magnetic wells, which recurs whenever the pitch angle becomes sufficiently large, and hence is inherently non-Markovian. This effect is more prominent for the lower- M_A case. Such a non-Markovian memory framework is essential for accurately modeling CR confinement, escape, and nonthermal emission in/around CR sources. In realistic astrophysical environments, such as the multiphase interstellar medium, M_A may vary over a wide range. Consequently, the balance between compressive mirror trapping and gyroresonant scattering is environment-dependent, leading to distinct CR transport phenomenology across different regions.

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Appendix A

Equivalence of Langevin Description and Focused Fokker–Planck Equation

From the Langevin model described in Equation (1),

$$\delta\mu = -\frac{1-\mu^2}{2B} v \cdot \nabla_s B \delta t + \frac{\partial D_{\mu\mu}}{\partial\mu} \delta t + \sqrt{2D_{\mu\mu}} \eta_r.$$

Here, $A = -\frac{1-\mu^2}{2B} v \cdot \nabla_s B + \frac{\partial D_{\mu\mu}}{\partial\mu}$, where s denotes the direction along the local magnetic field, and taking the

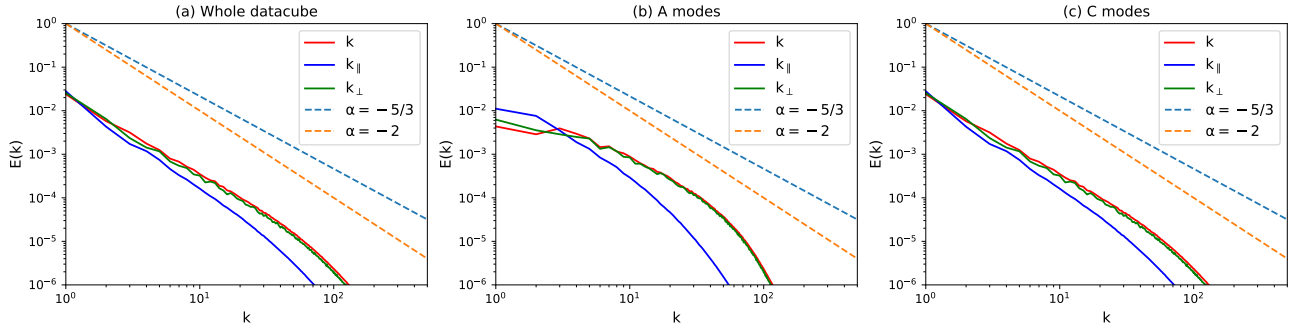


Figure 5. Power spectrum density (PSD) for the (a) whole datacube, (b) decomposed Alfvénic modes, and (c) decomposed compressible modes.

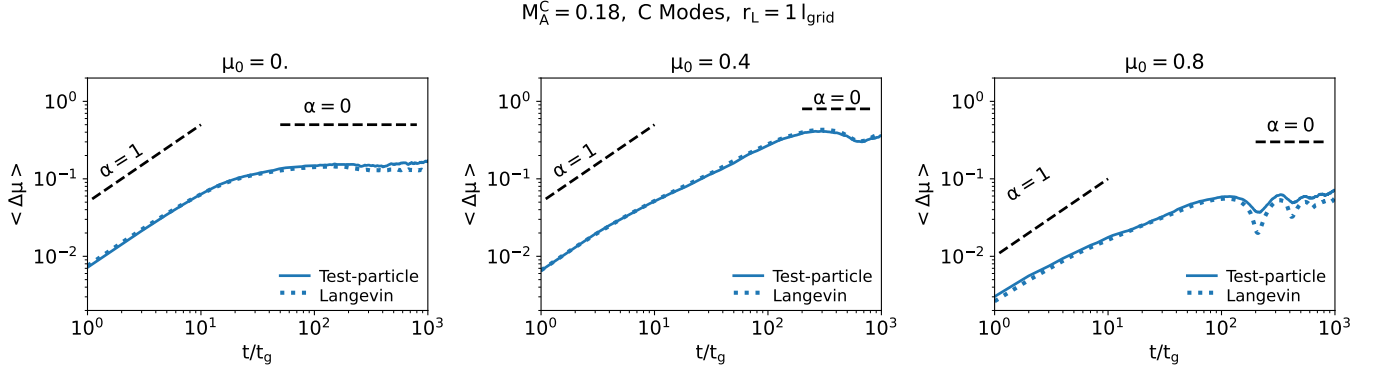


Figure 6. Time evolution of the mean pitch-angle increment $\langle \Delta \mu \rangle$ for particles with different initial μ_0 in a sub-Alfvénic compressible run ($M_A^C = 0.18$) with $r_L = 1 l_{\text{grid}}$. With an extremely small gyroradius, gyroresonance is suppressed in this case.

ensemble average, $\langle \delta \mu \rangle = A \delta t$, $\langle (\delta \mu)^2 \rangle = 2D_{\mu\mu} \delta t$. The probability transfer of the distribution function over a small time step gives

$$f(\mu, s, t + \delta t) = \langle f(\mu - \delta \mu, s - \delta s, t) \rangle.$$

We perform a Taylor expansion $f(\mu, s, t)$ about (μ, s) up to second order in $\delta \mu$ and first order in δs ,

$$f(\mu - \delta \mu, s - \delta s, t) = f(\mu, s, t) - \delta \mu \partial_\mu f(\mu, s, t) - \delta s \partial_s f(\mu, s, t) + \frac{1}{2} (\delta \mu)^2 \partial_\mu^2 f(\mu, s, t) + O(\delta t^{3/2}),$$

which leads to

$$f(\mu, s, t + \delta t) - f(\mu, s, t) = -\partial_\mu (A f) \delta t + \partial_\mu^2 (D_{\mu\mu} f) \delta t - v \mu \partial_s f \delta t.$$

Dividing by δt , we obtain

$$\begin{aligned} \frac{\partial f}{\partial t} + v \mu \frac{\partial f}{\partial s} &= -\frac{\partial}{\partial \mu} (A f) + \frac{\partial^2}{\partial \mu^2} (D_{\mu\mu} f) \\ &= \frac{\partial}{\partial \mu} \left[v(1 - \mu^2) \frac{\nabla_s B}{2B} f + D_{\mu\mu}(\mu) \frac{\partial f}{\partial \mu} \right]. \end{aligned}$$

This conserves the number of particles by integrating over all s and μ . By introducing the substitution $f = g/B$, one obtains

$$\begin{aligned} \frac{1}{B} \frac{\partial g}{\partial t} + v \mu \left(\frac{1}{B} \frac{\partial g}{\partial s} - \frac{\nabla_s B}{B} \frac{g}{B} \right) \\ = \frac{\partial}{\partial \mu} \left[v(1 - \mu^2) \frac{\nabla_s B}{2B} \frac{g}{B} + D_{\mu\mu} \frac{1}{B} \frac{\partial g}{\partial \mu} \right], \end{aligned}$$

which reduces to the focused Fokker–Planck equation

$$\frac{\partial g}{\partial t} + v \mu \frac{\partial g}{\partial s} - v(1 - \mu^2) \frac{\nabla_s B}{2B} \frac{\partial g}{\partial \mu} = \frac{\partial}{\partial \mu} \left(D_{\mu\mu} \frac{\partial g}{\partial \mu} \right).$$

Figures 5 and 6 provide supplementary diagnostics for the results discussed in the main text. Figure 5 shows the turbulence power spectra, while Figure 6 shows the pitch-angle evolution when gyroresonance is suppressed.

Appendix B

Numerical Comparisons of the Langevin Description and Focused Fokker–Planck Equation in the 1D Case

We present a 1D numerical comparison of the Langevin description and the focused Fokker–Planck equation in Figure 7. Since solving the Fokker–Planck equation in full 3D MHD turbulence is computationally expensive, we adopt a simplified 1D setup defined along the magnetic field where both approaches can be computed efficiently and compared directly. We construct a set of synthetic, gradient-varied magnetic fields $B(z)$ by superposing Fourier modes with random phases and a spectrum $E(k) \propto k^{-2} \exp(-k/50)$, and evaluate the corresponding focusing term. Here, the k^{-2} dependence is adopted to represent a spectrum measured along the local magnetic-field direction (A. Beresnyak 2015), while the exponential factor is included as an effective high- k dissipation cutoff. To approximate gyroresonant scattering, we set input diffusion coefficients $D_{\mu\mu}^G = D_0(1 - \mu^2)|\mu|$. We then solve the focused FP equation numerically and the Langevin equation with the same $B(z)$ and $D_{\mu\mu}^G$. Two representative parameter sets are considered: $M_A = 0.17$ with $D_0 = 0.005 \Omega_0$, and $M_A = 0.52$ with $D_0 = 0.04 \Omega_0$, where the gyrofrequency

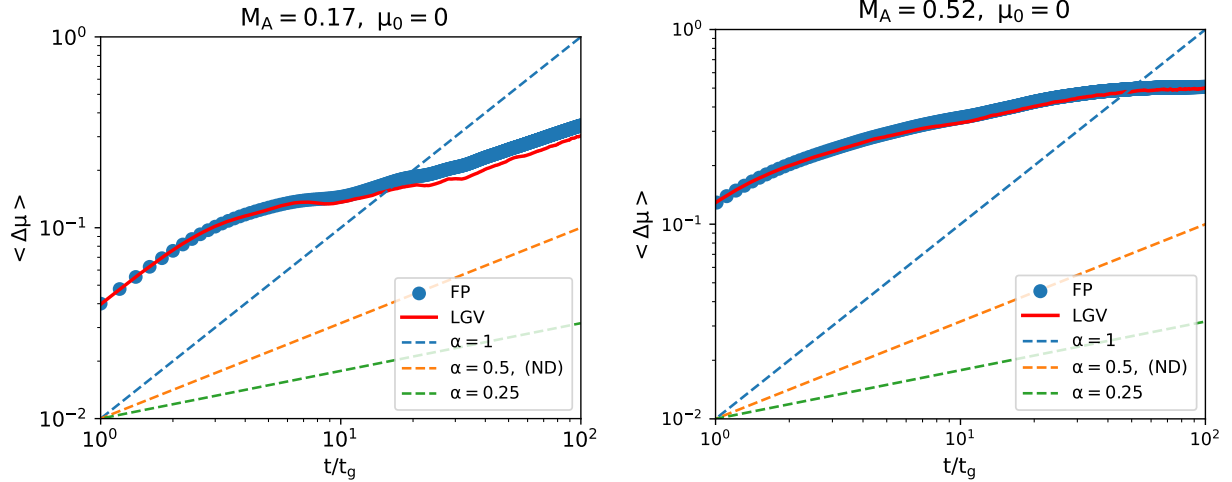


Figure 7. 1D numerical comparisons of focused Fokker-Planck (FP) solutions and Langevin solutions (LGV) under the low- M_A case (left panel) and the high- M_A case (right panel). $\alpha = 0.5$ represents the normal diffusion (ND) case.

Ω_0 is used to normalize the scattering rate. In both cases, the Langevin results are consistent with the FP solution, confirming the numerical equivalence of the two descriptions.

Appendix C Markovian Limit of the Mirror Interactions

If magnetic compressions experienced by the particles decorrelate sufficiently rapidly, mirror interactions can be coarse-grained into an effective stochastic process. In that limit, the relevant quantity is the pitch-angle diffusion coefficient $D_{\mu\mu}$, where (see R. Kubo 1957; A. Shalchi et al. 2004)

$$D_{\mu\mu}^{\text{Mirror}} = \int_0^\infty d\tau \langle \dot{\mu}_M(t) \dot{\mu}_M^*(t + \tau) \rangle.$$

Using

$$\dot{\mu}_M = -\frac{v(1 - \mu^2)}{2B_0} \nabla_{\parallel} \delta B_{\parallel},$$

one obtains

$$D_{\mu\mu}^{\text{Mirror}} = \frac{v^2(1 - \mu^2)^2}{4B_0^2} \int_0^\infty \langle \nabla_{\parallel} \delta B_{\parallel}(t) \nabla_{\parallel} \delta B_{\parallel}^*(t + \tau) \rangle d\tau,$$

after Fourier transform $\delta B_{\parallel}(\mathbf{x}, t) = \int \delta B_{\parallel}(\mathbf{k}, t) e^{i\mathbf{k} \cdot \mathbf{x}} d^3k$, $\nabla_{\parallel} \delta B_{\parallel} \rightarrow ik_{\parallel} \delta B_{\parallel}(\mathbf{k})$,

$$\begin{aligned} \langle \nabla_{\parallel} \delta B_{\parallel}(t) \nabla_{\parallel} \delta B_{\parallel}^*(t + \tau) \rangle &= \int d^3k d^3k' k_{\parallel} k'_{\parallel} \\ &\times \langle \delta B_{\parallel}(\mathbf{k}, t) \delta B_{\parallel}^*(\mathbf{k}', t + \tau) e^{i(\mathbf{k} - \mathbf{k}') \cdot \mathbf{x}} \rangle \\ &\simeq \int d^3k d^3k' k_{\parallel} k'_{\parallel} \langle \delta B_{\parallel}(\mathbf{k}, t) \delta B_{\parallel}^*(\mathbf{k}', t + \tau) \rangle \\ &\times \langle e^{i(\mathbf{k} - \mathbf{k}') \cdot \mathbf{x}} \rangle \\ &= \int d^3k d^3k' k_{\parallel} k'_{\parallel} P_{\parallel}(\mathbf{k}) \delta(\mathbf{k} - \mathbf{k}') \langle e^{i(\mathbf{k} - \mathbf{k}') \cdot \mathbf{x}} \rangle \\ &= \int d^3k k_{\parallel}^2 P_{\parallel}(\mathbf{k}) \langle e^{i\mathbf{k} \cdot \mathbf{x}} \rangle \end{aligned}$$

(see A. Shalchi et al. 2004). This leads to

$$D_{\mu\mu}^{\text{Mirror}} = \frac{v^2(1 - \mu^2)^2}{4B_0^2} \int d^3k k_{\parallel}^2 P_{\parallel}(\mathbf{k}) \int_0^\infty \langle e^{i\mathbf{k} \cdot \mathbf{x}} \rangle d\tau.$$

Consider that the GC has a Gaussian distribution along the field line, following H. Yan & A. Lazarian (2008),

$$\langle e^{i\mathbf{k} \cdot \mathbf{x}} \rangle = e^{i\mathbf{k}_{\parallel} \langle z \rangle} e^{-k_{\parallel}^2 \sigma_z^2 / 2}$$

$$\sigma_z^2 = \langle \Delta v_{\parallel}^2 \rangle t^2 = v_{\perp}^2 \left(\frac{\langle \delta B_{\parallel}^2 \rangle}{B_0^2} \right)^{1/2} t^2.$$

This gives the zeroth-order nonlinear broadened resonance function,

$$\begin{aligned} R_0 &= \int_0^\infty \langle e^{i\mathbf{k} \cdot \mathbf{x}} \rangle d\tau = \Re \int_0^\infty dt \\ &\times \exp \left[ik_{\parallel} v_{\parallel} t - \frac{1}{2} k_{\parallel}^2 v_{\perp}^2 t^2 \left(\frac{\langle \delta B_{\parallel}^2 \rangle}{B_0^2} \right)^{1/2} \right] \\ &= \frac{\sqrt{\pi}}{|k_{\parallel} \Delta v_{\parallel}|} \exp \left[-\frac{(k_{\parallel} v_{\parallel})^2}{k_{\parallel}^2 \Delta v_{\parallel}^2} \right] \\ &\simeq \frac{\sqrt{\pi}}{|k_{\parallel} v_{\perp} M_A^{1/2}|} \exp \left[-\frac{\mu^2}{(1 - \mu^2) M_A} \right]. \end{aligned}$$

Thus, we obtain the expression of the effective mirror-induced diffusion coefficient in Equation (6),

$$D_{\mu\mu}^{\text{Mirror}} = \frac{v^2(1 - \mu^2)^2}{4B_0^2} \int d^3k k_{\parallel}^2 P_{\parallel}(\mathbf{k}) R_0.$$

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